

EXECUTIVE SUMMARY

Predicting Combined Cycle Power Plant Electrical Output Using Machine Learning

A supervised regression pipeline for forecasting net hourly electrical energy output (PE) from ambient environmental conditions

Business Context & Objective

Combined Cycle Power Plants (CCPPs) generate electricity using paired gas and steam turbines, and their output fluctuates with ambient conditions rather than remaining constant. This project builds a machine learning model that predicts the plant's net hourly electrical energy output (PE, in megawatts) from four ambient measurements. Accurate short-horizon forecasts allow grid operators to plan power supply, avoid blackouts, and optimize trading on the spot energy market, while also supporting operational efficiency and emissions-reduction goals.

Modeling Task

This is a supervised regression problem: the target variable, PE, is a continuous numerical measurement in megawatts rather than a class label. The objective is to learn a function that maps four continuous ambient predictors to this continuous output, and to evaluate that function's accuracy on data it has not seen during training.

Dataset & Candidate Features

The pipeline uses 9,568 hourly average readings structured to mirror the well-known UCI CCPP dataset. No missing values were present. Four ambient variables serve as predictive features, each with a physically grounded relationship to power output:

Feature	Description	Relationship to Output (PE)
Ambient Temperature (AT)	Air temperature at the gas turbine intake (°C)	Strong inverse — higher AT lowers PE
Exhaust Vacuum (V)	Vacuum pressure at the steam turbine exhaust	Inverse — higher V lowers PE
Ambient Pressure (AP)	Atmospheric pressure (millibar)	Weak positive relationship with PE
Relative Humidity (RH)	Ambient relative humidity (%)	Mild inverse relationship with PE

Because these relationships are well documented in the thermodynamics of gas- and steam-turbine cycles (heat rejection efficiency drops as ambient temperature and humidity rise, and lower exhaust vacuum improves the steam cycle's Carnot efficiency), all four variables were retained as model inputs without further feature engineering.

Machine Learning Pipeline & Data Preprocessing

The workflow followed a standard, leakage-safe regression pipeline:

- Data integrity check — confirmed zero missing values and reviewed summary statistics (mean, standard deviation, min/max) across all five columns.
- Train/test split — the dataset was split 80/20 (7,654 training rows, 1,914 test rows) using a fixed random seed, with the 20% test set locked away as a final, unseen evaluation set ("test vault").
- Cross-validation strategy — a 5-fold cross-validation scheme (shuffled, fixed seed) was applied only to the training set, so every model comparison and hyperparameter choice was made without touching the test data.
- Hyperparameter tuning — GridSearchCV was used to tune the Ridge Regression regularization strength (α) across [0.01, 0.1, 1.0, 10.0, 100.0] using the same 5-fold splits.

Algorithms Evaluated

Two candidate algorithms, representing complementary modeling assumptions, were trained and compared:

- Random Forest Regressor — an ensemble of decision trees able to capture non-linear interactions between ambient variables without requiring feature scaling or an assumed functional form.
- Tuned Ridge Regression — a linear model with L2 regularization to control overfitting, with the penalty strength selected via grid search (best $\alpha = 100$).

Model selection was based on average cross-validation RMSE. Both models performed similarly, but the Random Forest Regressor produced the lower average error and was selected as the final model, then retrained on the full training set.

Model	Avg. Cross-Validation RMSE	Notes
Random Forest Regressor	7.874	Selected as final model (lowest CV error)
Tuned Ridge Regression ($\alpha = 100$)	7.887	Best linear alternative after grid search

Final Model Evaluation

The selected Random Forest model was evaluated once, on the held-out 20% test set that had not influenced any training or tuning decision. Results:

7.88 MW RMSE Typical error; penalizing large misses	6.06 MW MAE Average absolute prediction error	0.672 R² ~67% of output variance explained
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Interpretation & Business Implications

- Generalization: the cross-validation RMSE (7.874) and test-set RMSE (7.880) are nearly identical, indicating the model is not overfit and generalizes well to unseen conditions.

- Accuracy in context: an average error of roughly 6-8 MW against a mean output near 454 MW represents a small relative error, supporting the model's use in short-horizon operational forecasting.
- Explained variance: an R^2 of 0.672 means the four ambient features explain about two-thirds of the variation in output; the remaining third likely reflects factors outside the dataset (e.g., equipment condition, load demand, fuel characteristics, or maintenance state) that would need to be incorporated to further improve accuracy.
- Model choice: the Random Forest's edge over tuned Ridge Regression suggests the true relationship between ambient conditions and output includes mild non-linearity or interaction effects that a purely linear model cannot fully capture.

Conclusion

This pipeline demonstrates a complete and methodologically sound approach to forecasting CCPP electrical output: a clearly framed regression task, physically grounded features, a leakage-safe train/validation/test workflow, systematic comparison of a linear and a non-linear algorithm, and transparent evaluation against an untouched test set. The resulting Random Forest model offers a solid operational baseline, with clear next steps — additional plant- and demand-side features, extended hyperparameter search, and periodic retraining — identified as the most promising paths to improving predictive accuracy further.